Skills Summary

Unit Rates from Fractions and Mixed Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — A rate when both quantities are fractions

Rule: A unit rate is one quantity per *one* of the other, so it is always a division.

$$\text{rate} = \frac{\text{amount}}{\text{time}} = \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Dividing by a fraction smaller than 1 makes the answer *larger* — that is the built-in check.

Example: A tap delivers $\frac{2}{3}$ of a litre in $\frac{1}{6}$ of a minute. Litres per minute?
Step 1. Per minute means litres divided by minutes, so I divide the amount by the time and turn the division into multiplication by the reciprocal.
Step 2. $\frac{2}{3} \div \frac{1}{6} = \frac{2}{3} \times \frac{6}{1} = \frac{12}{3}$. Rate: **4** litres per minute

Try it: A tap delivers $\frac{3}{5}$ of a litre in $\frac{1}{4}$ of a minute. Litres per minute?

check: $\mathbf{\frac{12}{5}}$ litres per minute

Skill 2 — Rates that start with a mixed number

Rule: Convert every mixed number to an improper fraction *before* dividing.

$$w\frac{p}{q} = \frac{w \times q + p}{q}$$

Dividing a mixed number part by part gives a wrong answer, because the whole and the fraction do not divide separately.

Example: A conveyor moves $1\frac{3}{4}$ metres in $\frac{1}{2}$ of a minute. Metres per minute?
Step 1. The mixed number cannot be divided as it stands, so I rewrite it as an improper fraction first and only then divide.
Step 2. $1\frac{3}{4} = \frac{7}{4}$, and $\frac{7}{4} \div \frac{1}{2} = \frac{7}{4} \times \frac{2}{1} = \frac{14}{4}$. Rate: $\mathbf{\frac{7}{2}}$ metres per minute

Try it: A conveyor moves $4\frac{1}{2}$ metres in $\frac{3}{4}$ of a minute. Metres per minute?

check: **6** metres per minute

Skill 3 — Comparing two rates fairly

Rule: Two deals can only be compared once both are expressed *per one* of the same thing. A bigger total, a smaller price, or a shorter time proves nothing on its own. Work out both unit rates, put them over a common denominator, and then compare the numerators.

Example: Which is the greater rate, $\frac{9}{4}$ per hour or $\frac{7}{3}$ per hour?
Step 1. Both are already unit rates, so the only remaining job is to compare two fractions — I rewrite them over a common denominator.
Step 2. Over 12: $\frac{9}{4} = \frac{27}{12}$ and $\frac{7}{3} = \frac{28}{12}$, so 27 twelfths against 28 twelfths. $\dfrac{9}{4} < \dfrac{7}{3}$

Try it: Which is the greater rate, $\frac{8}{3}$ per hour or $\frac{5}{2}$ per hour?

check: $\frac{8}{3} > \frac{5}{2}$

Skill 4 — Turning the rate the other way up

Rule: "How long for one whole thing" is the *reciprocal* question, so divide the time by the fraction completed.

$$\text{time for one whole} = \frac{\text{time taken}}{\text{fraction completed}}$$

Check the direction: finishing more than was done must take longer than the time already spent.

Example: A machine completes $\frac{3}{4}$ of a job in $\frac{1}{2}$ of an hour. How long for the whole job?
Step 1. The question wants hours per job rather than jobs per hour, so I divide the time by the fraction of the job that is done.
Step 2. $\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{4}{6}$, and $\frac{2}{3} > \frac{1}{2}$ as it must be. Whole job: $\boldsymbol{\frac{2}{3}}$ of an hour

Try it: A machine completes $\frac{2}{5}$ of a job in $\frac{1}{3}$ of an hour. How long for the whole job?

check: $\boldsymbol{\frac{5}{6}}$ of an hour